

Question	Answer	Marks
8(a)	wavelength associated with a moving particle	B1
8(b)	$\lambda = h / p$	C1
	$= (6.63 \times 10^{-34}) / (9.11 \times 10^{-31} \times 4.9 \times 10^7)$ $= 1.5 \times 10^{-11} \text{ m}$	A1
8(c)	similarity: <i>any one point from:</i> • same mass • same <u>magnitude</u> of charge • both leptons	B1
	difference: <i>any one point from:</i> • electron has negative charge, positron has positive charge • positron is anti-particle of electron • electron is a particle, positron is an anti-particle	B1
8(d)(i)	(pair) annihilation	B1
8(d)(ii)	their mass gets converted into energy	B1
	(their mass–energy) becomes the energy of the gamma photons	B1
8(d)(iii)	they travel in opposite directions to conserve momentum	B1
8(d)(iv)	kinetic energy = $\frac{1}{2} \times 9.11 \times 10^{-31} \times (4.9 \times 10^7)^2 = 1.1 \times 10^{-15} \text{ J}$	A1

Question	Answer	Marks
8(d)(v)	$E = mc^2$	C1
	$E = hc / \lambda$ or $E = hf$ and $c = f\lambda$	C1
	$(1.1 \times 10^{-15}) + (9.11 \times 10^{-31} \times (3.00 \times 10^8)^2) = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / \lambda$ $\lambda = 2.39 \times 10^{-12} \text{ m}$	A1

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